STUDY-UNI · FULL MARKING SCHEME

FE3020

Understanding Supply Chain Management

Institution	University College Cork
Examination session	Winter 2023
Paper number	Paper 1
Duration of paper	1.5 hours
Special requirements	Calculator required
Instructions to candidates	Answer two questions. All questions carry equal marks.
Questions covered in this scheme	Question 1, Question 2, Question 3 (all three, since candidates may choose any two)

Question 1

20% + 30% + 50% = 100%

(a) **20%**

Briefly describe the bullwhip effect and identify the two main characteristics of this phenomenon, with the use of a sketched diagram and brief narrative.

FULL MODEL ANSWER

Description. The bullwhip effect is the tendency for order variability to increase at each successive stage moving upstream through a supply chain — consumer, retailer, distributor, manufacturer, supplier — even when actual consumer demand is relatively stable. It arises because, without shared information, each tier must guess what is happening downstream. Guessing wrong leads to too much or too little inventory: if too much, firms hold off ordering until stock falls (making the supplier upstream think demand has fallen); if too little, firms place a rush order and over-order to avoid being caught short again (making the supplier think demand has risen). Each tier's forecast error becomes the input to the next tier's forecast, so the distortion compounds with every step upstream.

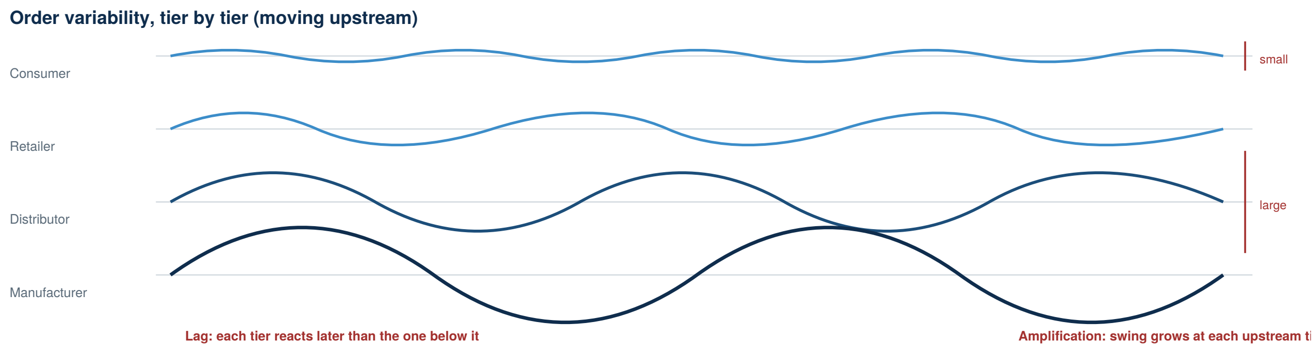

Figure 1 — Sketch of order variability by tier. The consumer's demand signal is relatively flat; by the time it reaches the manufacturer it has become larger in swing and later in timing.

Two main characteristics:

- **Amplification.** The size of the order swing grows at each upstream tier — a small, genuine change in consumer demand becomes a much larger swing in the orders a manufacturer or raw-material supplier sees.
- **Lag.** Each tier's response is delayed relative to the tier below it — information about a real change in demand takes time to travel upstream, so upstream tiers are reacting to yesterday's signal, not today's.

Narrative example. The module's "Green Volvo dilemma" illustrates both characteristics together: a mid-1990s spike in green Volvo sales was actually caused by dealers discounting unsold green cars to clear them off the lot, not by genuine demand growth. Because the manufacturer forecast from sales data rather than underlying orders, it read a lagged, amplified signal and dramatically increased green car production — the classic bullwhip pattern of over-reacting to a distorted, delayed demand signal.

Mark Allocation — Part (a) (worth 20% of the question)

Criterion	Marks	What earns the marks
Accurate description of the bullwhip effect	**8%**	Explains order-variability amplification moving upstream and the guessing/information-distortion mechanism
Diagram	**6%**	Sketch shows variability increasing (and/or lagging) across at least 3 tiers of the chain
Two characteristics correctly identified	**6%**	Amplification AND lag both named and correctly distinguished from each other
Total	**20%**	**Sum of criteria above**

(b) **30%**

Identify negative effects of the bullwhip on supply chain performance, specifying relevant metrics, and how these negative effects could be reduced.

FULL MODEL ANSWER

Negative effects and the metrics that capture them.

Negative effect	Relevant metric(s)
Excess inventory held to buffer against demand distortion	Inventory turns / days inventory on-hand
Storage, damage/pilferage, and obsolescence costs from that excess stock	Carrying cost component of total cost of ownership
Cash tied up funding the excess inventory	Cash-to-cash cycle time
Increased scheduling difficulty from volatile, distorted order patterns	Production plan adherence / schedule stability
Stock-outs when a rush order can't be filled fast enough	OTIF (perfect order) / service level

How these negative effects could be reduced:

- **Improve demand-stream visibility with real-time information sharing.** If genuine point-of-consumption demand is shared directly with upstream tiers, each tier plans against the real signal rather than a distorted, re-forecast version of it — directly reducing both amplification and lag.
- **Vendor Managed Inventory (VMI).** Handing replenishment control to the supplier, backed by a flow of real demand information, gives the supplier direct visibility over actual customer demand — the module identifies this explicitly as a control on the bullwhip effect — while also reducing buffer stocks and improving fill rates for the customer.
- **CPFR (Collaborative Planning, Forecasting and Replenishment).** Integrating multiple firms into a shared demand-planning process synchronises inventory strategies across the chain rather than leaving each tier to forecast in isolation.
- **Apply the inventory reduction principles** directly: pool inventory wherever demand can be combined, reduce variation at source, and reduce lead times — all three lower the safety stock a firm needs to hold to protect against the amplified, lagged signal.
- **Move toward pull/demand-driven replenishment** where lead times allow, so that downstream tiers trigger replenishment from actual consumption rather than push forecasts upstream in the first place.

Mark Allocation — Part (b) (worth 30% of the question)

Criterion	Marks	What earns the marks
Negative effects correctly identified	**10%**	At least three distinct effects (excess inventory/cost, cash tied up, scheduling difficulty, stock-outs)
Relevant metrics specified	**8%**	Each effect linked to a specific, correctly-named metric (inventory turns, cash-to-cash cycle, TCO, OTIF)
Reduction methods	**12%**	At least three distinct, correctly explained methods (e.g. information sharing, VMI, CPFR, inventory reduction principles, pull/demand-driven design)
Total	**30%**	**Sum of criteria above**

(c) **50%**

A large-scale consumer electronics manufacturing site has initiated a review of their direct goods (raw materials) buying policy. Currently buyers are paid a bonus if they achieve a price discount on purchases of 10% or higher. The buyers have taken advantage of the scale of the manufacturing site to place large orders with suppliers, sometimes enough in one delivery to satisfy site demand for an entire year.

Recently the business has launched an aggressive new product launch strategy to gain market share. The introduction of new models/upgrades requires retirement of older products.

The site manager has asked you to review the existing direct goods buying policy and introduce a purchasing and inventory management policy that reduces costs while maintaining material availability targets. He has also asked you to take into account rising interest rates.

As a starting point, you consider the total cost of ownership of materials. What are the main components of total cost of ownership for direct goods? How would you use this concept when devising a policy proposal for the site manager? Your answer should also address the immediate consequences of the current buying practice.

FULL MODEL ANSWER

Components of total cost of ownership for direct goods. The module's framework breaks TCO for a direct good (raw material) into:

- **Purchase price** — the unit price agreed with the supplier.
- **Cost of acquisition** — the transactional cost of placing and receiving the order.
- **Carrying costs**, comprising:
  - Material handling
  - Warehouse storage costs
  - Obsolescence & shrinkage
  - Insurance
  - Cost of capital

TCO component	What the current buying practice does to it
Purchase price	Minimised — this is the only line the bonus scheme actually optimises
Cost of acquisition	Reduced per unit (fewer, larger orders) but this is a small share of TCO
Material handling / warehouse storage	Increased — a full year's stock has to be received, moved and stored at once
Obsolescence & shrinkage	Sharply increased — a year's raw material stock is exposed to the new aggressive launch/retirement cycle
Insurance	Increased — insured value of stock on hand rises with a year's worth of material
Cost of capital	**Increased, and rising further — a year's stock is prepaid working capital, and interest rates are rising**

Immediate consequences of the current buying practice. The bonus scheme rewards buyers purely on the purchase-price line of TCO — a 10%+ discount — while ignoring every other component. Placing orders large enough to cover a full year of site demand is a rational response to that specific incentive, but it is a textbook case of local optimisation: it minimises purchase price at the direct expense of carrying cost. Three consequences follow immediately from the scenario as described:

- **Obsolescence risk has just increased sharply**, because the business has simultaneously launched an aggressive new product strategy that retires older models. Raw material bought a year ahead for a product that may be discontinued mid-cycle becomes stranded inventory — write-offs that can easily exceed the 10% discount that justified the order in the first place.
- **Cost of capital is rising at the same time as the practice ties up more of it** — a full year of prepaid raw material inventory is exactly the wrong exposure to be increasing as interest rates rise, since the financing cost of that stock (part of "carrying cost") grows in step with rates.
- **The incentive structure itself is the root cause**, not an isolated buyer decision — as long as buyer bonuses are tied only to purchase-price discount, buyers will continue to be rewarded for decisions that a full TCO view would show are net negative for the site.

> **SOURCING NOTE** *The framing of the bonus scheme as a 'local optimisation' problem — optimising one TCO line item at the expense of the others — is standard purchasing/procurement reasoning applied to this specific scenario; it is not a named model in the module slides, so it should be credited on the strength of the TCO-component reasoning above, not as recall of a named framework.*

Using TCO to devise the policy proposal. The core recommendation is to re-anchor the buying policy — and the buyer incentive that drives it — on total cost of ownership rather than purchase price alone:

- **Redesign the bonus metric** away from "discount achieved" toward a net TCO saving measure, so a smaller discount earned through a materially lower carrying-cost profile can score at least as well as today's headline 10%+ discount bonus.
- **Move from single annual bulk orders toward smaller, more frequent deliveries**, matched more closely to the new, faster product life cycles created by the aggressive

launch strategy — directly reducing the warehouse storage, obsolescence, insurance and cost-of-capital lines, at the cost of a smaller purchase-price discount and a marginally higher per-order acquisition cost.

- **Explicitly price in rising interest rates** when comparing options: a 10% price discount funded by a year's prepaid inventory needs to be weighed against a cost of capital that is now higher and rising, which shrinks — and can reverse — the apparent saving the bonus scheme currently rewards.
- **Differentiate the policy by product/material risk**: materials tied to models the launch strategy is about to retire should carry the tightest ordering discipline (smallest order sizes, shortest coverage horizon), while materials common across current and next-generation models can safely support larger orders, since obsolescence risk is much lower for them.
- **Maintain material availability targets** by using TCO-informed safety stock (driven by supplier lead time and criticality) rather than bulk-buying as the mechanism for availability — availability and low unit cost do not have to be pursued through the same lever.

Mark Allocation — Part (c) (worth 50% of the question)

Criterion	Marks	What earns the marks
TCO components for direct goods correctly listed	**12%**	Purchase price, cost of acquisition, and all five carrying-cost elements identified
Immediate consequences of current buying practice	**13%**	Explicitly connects large one-off orders to obsolescence risk (via the new launch strategy) AND rising cost of capital (via rising interest rates)
TCO used to devise the policy	**15%**	Concrete, TCO-anchored policy proposals (incentive redesign, order-frequency change, risk-differentiated ordering) rather than generic cost-cutting advice
Addresses material availability target explicitly	**5%**	Shows the proposal still protects availability, not just cost
Coherence — TCO used as the organising concept throughout	**5%**	Answer is structured around TCO rather than treating it as a single opening definition
Total	**50%**	**Sum of criteria above**

Question 1 — total	100%

Question 2

60% + 40% = 100%

(a) **60%**

Supply chain development has been characterised by specialisation and outsourcing. Using the example of grocery supply chains, illustrate and discuss the supply chain design before and after the introduction of central distribution, highlighting performance implications. Your answer should include basic before and after supply chain design schematics and identify key metric(s) when discussing performance.

FULL MODEL ANSWER

Specialisation and outsourcing framing. Central distribution is itself an example of the specialisation and outsourcing trend: rather than every supplier managing its own delivery relationship with every store, distribution and consolidation is concentrated into a specialist function (frequently an appointed 3PL), and each party specialises in the part of the chain it does best — suppliers in production, the 3PL in distribution, the retailer in store operations.

Before central distribution. Consumer food processors deliver directly to individual retail stores, typically limited to around twice-weekly delivery. Each store holds and manages its own inventory independently — there is no pooling of stock across the retailer's network, and delivery frequency is constrained by the economics of visiting many individual stores directly.

After central distribution. Delivery shifts to daily supplier deliveries into a central (or regional) distribution centre, with onward daily delivery to retail stores driven by daily store orders. Logistics arrangements range from delivery into the RDC to backhaul by the retailer's own distribution fleet from the supplier's factory gate (Factory Gate Pricing), and RDC operations frequently involve cross-docking — immediate despatch of supplier orders from the RDC to stores without being taken into stock — often run by an appointed 3PL.

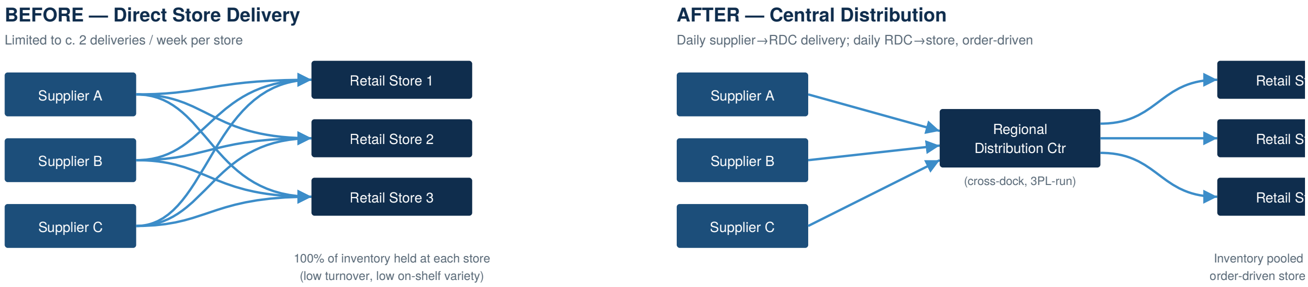

Figure 1 — Grocery supply chain design before and after the introduction of central distribution. Arrows indicate physical product flow; inventory pools at store level before, and at the RDC after.

Performance implications:

- **Reduced days inventory on-hand throughout the chain** — pooling stock at the RDC rather than fragmenting it across every individual store lowers total inventory required for the same service level, and cross-docking further reduces RDC dwell time.

- **Improved information flow, reducing bullwhip** — daily, order-driven replenishment gives much tighter, higher-frequency demand signals back up the chain than twice-weekly direct delivery, reducing the amplification and lag that drive the bullwhip effect.
- **Enhanced stock turnover and on-shelf variety at retail** — more frequent, smaller, order-driven deliveries let stores carry a wider range without a proportional increase in backroom stock.
- **Requires continuous replenishment capability and high supplier compliance** — daily, order-driven delivery only works if suppliers reliably deliver on time and in full; this is why OTIF becomes a key operating metric under this model.
- **Opportunity and dependency for suppliers** — closer, higher-frequency integration with a large retailer creates the basis for a category-leadership relationship, though it also concentrates dependency on that retailer's central distribution requirements — a direct consequence of the specialisation trend the question asks you to frame this within.

Key metrics. The module identifies inventory turnover, the cash-to-cash cycle time, and (retailer-specific) sales per m^2 as the core metrics for assessing the performance implications of this shift, alongside OTIF as the compliance metric that continuous replenishment depends on.

Mark Allocation — Part (a) (worth 60% of the question)

Criterion	Marks	What earns the marks
Frames the answer around specialisation/outsourcing	**5%**	Explicitly connects central distribution to the specialisation/outsourcing trend named in the question
Accurately describes the 'before' design	**10%**	Direct-to-store delivery, twice-weekly, store-level inventory correctly identified
Accurately describes the 'after' design	**10%**	Daily RDC-based delivery, cross-docking, 3PL/backhaul correctly identified
Schematic diagrams provided (before and after)	**10%**	Basic before/after schematics included and broadly correct in structure
Performance implications discussed	**15%**	At least three implications discussed with correct reasoning
Key metric(s) correctly identified	**10%**	Inventory turnover AND cash-to-cash cycle AND/OR sales per m^2 AND OTIF, correctly tied to the discussion
Total	**60%**	**Sum of criteria above**

(b) **40%**

CK Foods produces a range of soups and 'hot pot meals' for the food service market. Their point of differentiation is the high level of customisation offered to their customers, enabling them to continually update their menus. This service commands a premium price. Over a number of years, they have built strong relationships with a range of specialist food ingredient suppliers, especially for seasoning and flavouring such as herbs, spices, fruit zests, essences and oils. Many of these suppliers are small scale, however the source integrity they offer is of fundamental importance. This supplier base responds quickly to orders which enables CK Foods to use a make-to-order fulfilment strategy.

The business was acquired last year by a large multinational corporation. Finance in Corporate HQ have requested implementation of a 'days payable policy' of 60 days for raw material suppliers. Given their small-scale, currently CK Foods offer these suppliers a 'days payable' period of 30 days.

Currently CK Foods holds inventory on hand for 30 days on average, and its customers pay within 60 days. Calculate the cash-to-cash cycle time? What concerns would you have about this change in terms of trade with suppliers? How would you respond to the request for Finance at Corporate HQ?

FULL MODEL ANSWER

Cash-to-cash cycle formula (module content): Days Inventory On-hand + Days Sales Outstanding − Days Payables Outstanding = Cash-to-cash days.

Scenario	DIO	DSO	DPO	Cash-to-cash
Current (30-day supplier terms)	30	60	30	60 days
Proposed (Group 60-day policy)	**30**	**60**	**60**	**30 days**

30 + 60 − 30 = **60 days** currently. 30 + 60 − 60 = **30 days** under the Group's proposed policy — the cycle would **halve**, a 30-day reduction, which releases a substantial amount of cash and is numerically very attractive from a Group cash-conversion perspective.

Verdict — Numerically the strongest possible case for Finance, but the most dangerous version of this trade-off for CK Foods' supplier base

Concerns about this change in terms of trade. This is a more severe version of the trade-off than a modest extension: the request is to *double* supplier payment terms, from 30 to 60 days, not extend them incrementally. CK Foods' entire competitive proposition — continual menu customisation, make-to-order fulfilment, premium pricing — is built on small-scale speciality suppliers (herbs, spices, fruit zests, essences, oils) who are explicitly described as responding quickly to orders, which by construction means they are carrying minimal inventory and minimal financial buffer themselves. Doubling their payment period is a direct cashflow shock to exactly the suppliers CK Foods can least afford to destabilise: if even a handful of them face financial difficulty or stop prioritising CK Foods' orders, the make-to-order strategy and the "source integrity" that justifies the premium price both break down simultaneously.

How to respond to Corporate HQ. As with any cash-to-cash policy change imposed from Group level, the response should not be a flat refusal or a silent roll-out, but a differentiated implementation informed by supplier profiling:

- **Segment the supplier base before applying new terms** — apply 60-day terms only to suppliers with the financial scale to absorb it, and protect the small-scale speciality suppliers whose responsiveness is the actual source of CK Foods' premium positioning.
- **Offer supply chain finance / early-payment facilities** for the suppliers where 60-day terms would otherwise threaten continuity — the supplier is paid promptly via a financing arrangement while CK Foods still books the 60-day term for Group cash-to-cash purposes.
- **Make the integration risk explicit to HQ**: CK Foods was acquired specifically for a differentiated, premium proposition built on this supplier network; a blanket Group-wide payables policy applied without reference to that network risks eroding the very asset the acquisition was made for. This needs to be surfaced as a named risk to Corporate HQ, not absorbed quietly at site level.

Mark Allocation — Part (b) (worth 40% of the question)

Criterion	Marks	What earns the marks
Correct formula applied	**4%**	DIO + DSO − DPO stated and applied correctly
Correct current cash-to-cash calculation	**4%**	30 + 60 − 30 = 60 days, shown with full working
Correct proposed cash-to-cash calculation	**4%**	30 + 60 − 60 = 30 days, shown with full working, and the 30-day change correctly identified
Concerns specific to CK Foods' supplier base	**13%**	Explicitly connects the risk to small-scale speciality suppliers and the make-to-order/source-integrity proposition, not a generic answer
Recognises the severity relative to a smaller extension	**5%**	Notes that doubling (not just extending) terms is a materially larger shock
Proposes a workable response to HQ	**10%**	At least one concrete mechanism (segmentation, supply chain finance, escalation of acquisition/integration risk)
Total	**40%**	**Sum of criteria above**

Question 2 — total	**100%**

Question 3

20% + 20% + 30% + 40% printed = 110%

SOURCING NOTE *The source paper's instruction line reads 'Answer parts (a), (b) and (c)', but a part (d) — the MH Foods inventory turnover scenario — is also printed with its own 40% allocation. The four printed weights (20% + 20% + 30% + 40%) sum to 110%, not 100%. This is an inconsistency in the source paper itself, most likely from the paper being adapted from an earlier version without every figure being re-checked — it is flagged here rather than silently rescaled. All four parts are marked below at the percentages actually printed on the paper, since all four are genuine, answerable content regardless of which three the instruction line intends.*

(a) **20%**

Describe three types of logistics providers that a company may use to get product to market.

FULL MODEL ANSWER

- **Hauliers.** Most appropriate where both the company and its customer/distributor already have their own warehouses and storage facilities, so goods can be picked, packed and despatched from one facility to an equivalent facility for onward distribution. The emphasis is on a point-to-point transport provider, without wider logistics management.
- **Forwarders (freight forwarders).** Facilitate the movement of goods around the world on behalf of importers and exporters — handling customs, preparing and completing documentation, finding the best cost/speed routes, and buying and selling space on ships and aircraft. Their role is coordination and documentation rather than owning the physical transport asset.
- **Third Party Logistics providers (3PL).** Manage all aspects of a company's distribution from one end of the supply chain to the other, using their own asset-based resources (infrastructure and transport) and often a network of partners — effectively combining the functions of both a forwarder and a transport provider under a single, broader logistics management relationship.

Mark Allocation — Part (a) (worth 20% of the question)

Criterion	Marks	What earns the marks
Hauliers correctly described	**6%**	Point-to-point transport between the company's and customer's own facilities
Forwarders correctly described	**7%**	Customs, documentation, and route/space brokerage on behalf of importers/exporters
3PL correctly described	**7%**	Combines forwarder and transport functions, own assets plus partner network, manages the full distribution chain
Total	**20%**	**Sum of criteria above**

(b) **20%**

List four advantages and four disadvantages of outsourcing logistics.

FULL MODEL ANSWER

#	Advantages of outsourcing	Disadvantages of outsourcing
1	Increased capability to handle volume, complexity, and geographic reach, including new product and market launches	Loss of direct control over the customer relationship
2	Reduced operating cost through economies of scale and shared infrastructure	Reduced control of customer service — order accuracy, presentation, and delivery time
3	Frees up capital and management attention to focus on core activities; assets removed from the balance sheet	Risk of losing specific company/industry knowledge that would otherwise sit in-house
4	Risk management and access to expertise — commercial and HR risk spread across operators, and best practice gained from the 3PL's broader client base	Reduced flexibility from long-term contractual commitments, plus an added operator margin on top of the underlying cost

Mark Allocation — Part (b) (worth 20% of the question)

Criterion	Marks	What earns the marks
Four advantages, correctly explained	**10%**	Four distinct advantages given, each with a brief correct rationale (2.5% each)
Four disadvantages, correctly explained	**10%**	Four distinct disadvantages given, each with a brief correct rationale (2.5% each)
Total	**20%**	**Sum of criteria above**

(c) **30%**

The 'True Cost' concept, as popularised by Andrew Morgan (2015), has sparked considerable debate about 'fast fashion'. Identify two key sustainability issues highlighted by this concept and how the 'True Cost' approach seeks to address them. Your answer should explicitly address actions to take account of 'externalities' identified by a true cost assessment.

FULL MODEL ANSWER

SOURCING NOTE *Andrew Morgan's 2015 documentary* The True Cost *and its specific application to fast fashion are not named in the module slides, which present the True Cost / externalities framework primarily through food-system sources (the True Price Foundation and the Rockefeller Foundation's True Cost of Food work). This answer applies the module's own True Cost and externalities framework to the fast fashion context the question specifies; the biographical/documentary detail was checked independently and the framework content is the module's own.*

The concept. True cost economics represents the gap between a product's market price and the total societal cost of producing it — including negative externalities such as environmental harm and impacts on public health that are not reflected in the price paid. *The True Cost* (2015), directed by Andrew Morgan, applied this lens to the fast fashion industry, following the supply chain behind low-cost, rapidly-cycled clothing back to its garment-worker and environmental impacts, prompted in part by the 2013 Rana Plaza factory collapse in Bangladesh.

Two key sustainability issues:

- **Human rights / labour conditions (social externality).** Fast fashion's low retail prices are enabled in part by unsafe factory conditions and very low wages in garment-producing countries — a cost borne by workers rather than reflected in the price consumers pay. This maps directly onto the module's "social" category of externality: underpayment, poor working conditions, and health and safety risk sitting outside the transaction price entirely.
- **Environmental damage (environmental externality).** The fast fashion production cycle — pesticide-intensive cotton farming, dye and chemical pollution of water and soil, and a linear take-make-waste model that ends in landfill — imposes environmental costs on producing regions and future consumers that are likewise absent from the shelf price. This maps onto the module's "environmental" externality category: water and soil pollution, overuse of resources, and contribution to a linear rather than circular model.

How the True Cost approach seeks to address them. Rather than treating these as unpriced side-effects, a true cost assessment quantifies the gap between market price and true societal cost for a given garment, making the externality visible and therefore actionable — shifting sustainability from the peripheral "true cost/externalities" discussion into something that can be structurally priced, reported, and managed, in the same way the module frames ESG's ambition for supply chains more broadly.

Actions to take account of externalities (module framework, applied to fast fashion):

- **Creating business and consumer preference for lower true-price-gap products** — e.g. favouring garments from factories with verified labour standards and lower-impact material sourcing.
- **Enabling businesses to develop strategies that reduce their own true-price gap** — for example redesigning supply chains to cut pollution intensity or improve factory safety and wages.
- **Enabling businesses to directly take up remediation activities** — restoring or compensating affected workers or communities at full value where harm has occurred, taking steps to prevent recurrence, and cooperating with governments on breaches of labour or environmental obligations.
- **Enabling consumers to pay the true-price gap on top of the transaction price**, where their means allow — e.g. a premium price tier that funds remediation directly.
- **Shaping government policy** to incentivise sustainable and inclusive decision-making across the fashion supply chain — regulation, disclosure requirements, or trade conditions tied to labour and environmental standards.

Mark Allocation — Part (c) (worth 30% of the question)

Criterion	Marks	What earns the marks
Two key sustainability issues correctly identified	**10%**	Human rights/labour AND environmental damage, each correctly linked to fast fashion specifically
Explains how True Cost approach addresses them	**8%**	Explains the market-price vs societal-cost gap and how quantifying it makes the issue actionable
Actions to account for externalities	**10%**	At least three of the module's specific proposed actions correctly applied to the fast fashion context
Coherence and grounding in the True Cost framework	**2%**	Answer stays anchored to the True Cost/externalities concept rather than a generic sustainability answer
Total	**30%**	**Sum of criteria above**

(d) **40%**

MH Foods is a well-established business and for many years supplied a generic portfolio of frozen desserts to a range of food service customers. Last year they hired a new salesperson who has increased the number of customers. This has been achieved by offering customised products, including various new ingredients often from new suppliers. MH Foods pride themselves on quick design and delivery of this increasing variety of products. This has boosted sales, but Finance have reported an erosion of profit margin. Furthermore, the warehouse manager has rented additional space due to the range of ingredients required and less frequent delivery by new suppliers of bespoke ingredients.

The Managing Director has asked Finance for key numbers so she can have a better sense of the current situation. They have given her the following. In 2020, the total cost of goods sold was €20 million, while average inventory holding was €2 million. At the end of 2021 sales increased and the cost of goods sold was €24 million, while average inventory was €3 million.

Using the formula for inventory turnover, calculate the inventory turnover for this scenario. Comment on the performance achieved, including identification of possible reasons for this and measures that might be taken to improve performance.

FULL MODEL ANSWER

Formula (module content): Inventory Turnover = Cost of Goods Sold ÷ Average Inventory Held. The higher the turnover, the lower the firm's inventory costs, for a given level of sales.

Year	COGS	Avg. inventory	Turnover	≈ Days inventory
2020	€20m	€2m	10×	36.5
2021	**€24m**	**€3m**	**8×**	**45.6**

2020: €20m ÷ €2m = **10×**. 2021: €24m ÷ €3m = **8×**. Turnover fell from 10× to 8×, equivalent to inventory rising from roughly 36.5 days on hand to roughly 45.6 days.

Verdict — Inventory turnover performance has declined, despite rising sales

Comment on the performance achieved. This looks, at first glance, like a success story — COGS grew 20% year-on-year, consistent with the new salesperson's growth in customer numbers. But average inventory grew faster still, by 50%, so the business is now carrying proportionally far more stock to support the same revenue base than it was the year before. Turnover falling is a genuine performance decline, not just a side-effect of growth — growing sales and falling inventory efficiency are two separate outcomes that have both occurred here, and the second one is quietly eroding the value created by the first.

Possible reasons for this. The scenario itself supplies the two most likely drivers:

- **Rising product variety from customisation.** A widening range of bespoke ingredients, each often used for only a subset of customers' menus, increases the total inventory needed to support the same sales base — many of these ingredients carry low commonality with each other, so pooling and substitution benefits that a generic portfolio would have are largely lost.
- **Less frequent delivery from the new bespoke-ingredient suppliers**, stated directly in the scenario — infrequent delivery from these new, smaller suppliers forces MH Foods to hold larger buffer stocks of ingredients that may only be used in exceptional or lower-volume orders, increasing the risk that some of that inventory becomes obsolete or damaged before use, which is itself consistent with the reported profit margin erosion (write-offs and holding cost directly reduce margin).
- The additional warehouse space rented is the physical, costed evidence of this inventory build-up — not a separate cause, but a direct consequence of the same underlying driver.

Measures to improve performance.

- **Monetise the cost of the additional inventory** (holding cost, the incremental warehouse rent, and any obsolescence/write-off exposure) and share this explicitly with the sales team, so the trade-off between customer-count growth and inventory-carrying cost is visible at a decision-making level, not just in the year-end margin numbers.
- **Review pricing for highly customised orders** so that the true cost of carrying low-volume, bespoke ingredient inventory is reflected in what those specific customers are charged, rather than being absorbed across the whole product range.
- **Renegotiate lead times and delivery frequency with the new suppliers**, moving toward smaller, more frequent deliveries to reduce the buffer stock currently required to cover infrequent delivery windows.
- **Consider postponement or quick-changeover techniques** as the business scales, so that final customisation happens later and closer to the order, reducing the amount of bespoke inventory that has to be held speculatively in advance.
- **Track lead-time and total cost of ownership alongside inventory turnover** going forward, since — as with the earlier CK/CW Foods-style scenarios in this paper — inventory turnover alone doesn't show why performance has moved, only that it has.

Mark Allocation — Part (d) (worth 40% of the question)

Criterion	Marks	What earns the marks
Correct formula and 2020 calculation	**6%**	Turnover = COGS/Avg Inventory; 20/2 = 10x shown with working
Correct 2021 calculation	**6%**	24/3 = 8x shown with working
Correctly identifies that performance has worsened	**6%**	States clearly that turnover fell (not just that both numbers changed) despite COGS growth
Reasons for the decline	**12%**	Identifies product-variety/customisation driver AND the stated less-frequent-delivery driver, linked to the margin erosion
Improvement measures	**10%**	At least three specific, relevant measures (e.g. cost visibility to sales, pricing review, delivery renegotiation, postponement)
Total	**40%**	**Sum of criteria above**

Question 3 — total (as printed on the paper) 110%